Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)	I	$2Cr^{3+} + 3H_2O_2 + H_2O \rightarrow Cr_2O_7^{2-} + 8H^+$		1		1		
			II	reaction is feasible (must give reason to gain this mark) (1) as standard electrode potential for hydrogen peroxide is more positive than that for chromium/dichromate / EMF calculated as +0.44V and positive value means reaction is feasible (1)		1	1	2		
			III	electrochemical methods are better (must give reason for this mark) (1) because the reaction is in solution but standard enthalpies of formation use standard states			2	2		
				Question 11 total	**3**	**7**	**8**	**18**	**4**	**3**